

Quantum Tunneling

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Let a particle or a wave packet (in the language of quantum mechanics) of fixed mass m and fixed energy E travel along the x -direction (from left to right). It encounters a potential energy barrier $V(x)$. In this scenario, the time-independent Schrödinger equation applies, and the wavefunction $\psi(x, t)$ (where t is the time) is factored into its spatial part $\Psi(x)$, which satisfies:

$$-\frac{\hbar^2}{2m} \frac{d^2\Psi(x)}{dx^2} + V(x)\Psi(x) = E\Psi(x) \quad (1)$$

where:

- $\hbar$: Reduced Planck constant.
- m : Mass of the particle/wave packet.
- $\Psi(x)$: Wavefunction, representing the probability density ($|\Psi(x)|^2$) of finding the particle at position x .
- $\frac{d^2\Psi(x)}{dx^2}$: Second spatial derivative of the wave function related to the kinetic energy.
- $V(x)$: Position-dependent potential energy.
- E : Total energy of the particle/wave packet.

This form assumes that the system is time independent, meaning that the energy E is constant and the time evolution of the wavefunction has been factored out. The spatial part $\Psi(x)$ satisfies the stationary Schrödinger equation.

The potential profile is defined as:

$$V(x) = \begin{cases} 0 & \text{for } x < 0 \quad (\text{Region I}) \\ V_0 & \text{for } 0 \leq x \leq L \quad (\text{Region II}) \\ 0 & \text{for } x > L \quad (\text{Region III}) \end{cases} \quad (2)$$

We also consider $V_0 > E$.

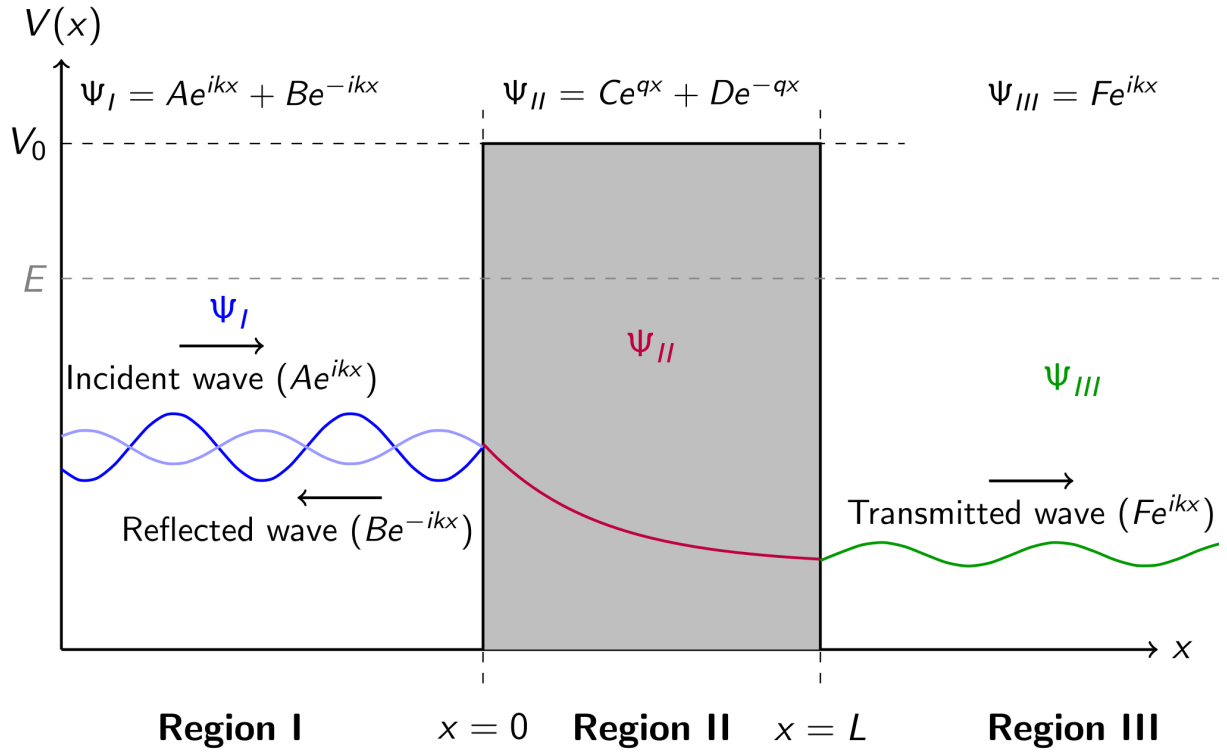


Figure 1: Wavefunction behavior across a potential barrier.

General Solution

We consider the second-order differential equation:

$$\frac{d^2 y}{dx^2} = a^2 y \quad (3)$$

We try a solution of the form $y(x) = e^{bx}$. Substituting into the equation:

$$\frac{d^2}{dx^2} e^{bx} = b^2 e^{bx} = a^2 e^{bx} \Rightarrow b^2 = a^2 \quad (4)$$

This yields two cases:

- If $a^2 > 0$, then $b = \pm a$, and the general solution is:

$$y(x) = c_1 e^{ax} + c_2 e^{-ax} \quad (5)$$

- If $a^2 < 0$, define $d = \sqrt{-a^2}$ so that $b = \pm id$, and the general solution becomes:

$$y(x) = c_1 e^{idx} + c_2 e^{-idx} \quad (6)$$

Region I ($x < 0$): Free space

In this region, the potential is zero: $V(x) = 0$. The time-independent Schrödinger equation becomes:

$$-\frac{\hbar^2}{2m} \frac{d^2 \Psi_I(x)}{dx^2} = E \Psi_I(x) \quad (7)$$

Rewriting in the form $\frac{d^2 y}{dx^2} = a^2 y$, we get:

$$\frac{d^2 \Psi_I(x)}{dx^2} = -\frac{2mE}{\hbar^2} \Psi_I(x) \quad (8)$$

This corresponds to oscillatory solutions since the coefficient is negative. Define:

$$k = \frac{\sqrt{2mE}}{\hbar} \quad (9)$$

The general solution is:

$$\Psi_I(x) = Ae^{ikx} + Be^{-ikx} \quad (10)$$

The term Ae^{ikx} represents the incident wave traveling in the positive x direction (from left to right), with wave number k corresponding to the momentum of the particle in free space. This component describes the incoming probability amplitude as it approaches the potential barrier.

In contrast, Be^{-ikx} denotes the wave reflected propagating in the negative x -direction (right to left), arising due to partial reflection at the boundary $x = 0$. Its presence indicates that the barrier is not fully transparent to the incoming wave.

We will see later that the squared magnitudes $|A|^2$ and $|B|^2$ can be related to the probability current densities associated with the incident and reflected waves, respectively. The ratio

$$R = \frac{|B|^2}{|A|^2} \quad (11)$$

defines the reflection coefficient, quantifying the fraction of the incoming wave that is reflected by the barrier.

Region II ($0 \leq x \leq L$): Barrier region

Here, the potential is constant: $V(x) = V_0$. The Schrödinger equation is:

$$-\frac{\hbar^2}{2m} \frac{d^2 \Psi_{II}(x)}{dx^2} + V_0 \Psi_{II}(x) = E \Psi_{II}(x) \quad (12)$$

Rewriting:

$$\frac{d^2 \Psi_{II}(x)}{dx^2} = \frac{2m(V_0 - E)}{\hbar^2} \Psi_{II}(x) \quad (13)$$

This corresponds to exponential solutions since $V_0 > E$. Define:

$$q = \frac{\sqrt{2m(V_0 - E)}}{\hbar} \quad (14)$$

The general solution is:

$$\Psi_{II}(x) = Ce^{qx} + De^{-qx} \quad (15)$$

The general solution in Region II consists of two exponential terms: Ce^{qx} and De^{-qx} . These do not represent traveling waves but rather spatially growing and decaying components, respectively, since q is real and positive for $E < V_0$.

The term Ce^{qx} grows with x , while De^{-qx} decays with x . However, physical considerations typically require that the wave function remain finite within the barrier, especially near $x = L$, so the relative weights of C and D are determined by boundary conditions in $x = 0$ and $x = L$.

This exponential behavior reflects the quantum tunneling effect: although the particle does not have enough classical energy to overcome the barrier, the wavefunction penetrates the classically forbidden region, resulting in a nonzero probability density amplitude inside the barrier.

The decay constant q quantifies the rate at which the wave function attenuates within the barrier.

Region III ($x > L$): Free space again

In this region, the potential returns to zero: $V(x) = 0$. The time-independent Schrödinger equation becomes:

$$-\frac{\hbar^2}{2m} \frac{d^2 \Psi_{III}(x)}{dx^2} = E \Psi_{III}(x) \quad (16)$$

Rewriting in the canonical form:

$$\frac{d^2 \Psi_{III}(x)}{dx^2} = -\frac{2mE}{\hbar^2} \Psi_{III}(x) \quad (17)$$

This again corresponds to oscillatory solutions, just like Region I. Using the same wave number k (see Eq. 9), the general solution is:

$$\Psi_{III}(x) = F e^{ikx} + G e^{-ikx} \quad (18)$$

Here, $F e^{ikx}$ represents a transmitted wave propagating in the positive x -direction (from left to right), beyond the barrier. The term $G e^{-ikx}$ would correspond to a wave traveling in the negative x direction (right to left), implying a source or reflection from infinity.

However, since Region III lies beyond the barrier and there is no physical mechanism to generate a wave from the far right, we impose the condition:

$$G = 0 \quad (19)$$

Thus, the physically acceptable solution in Region III is:

$$\Psi_{III}(x) = F e^{ikx} \quad (20)$$

Derivatives of the Wavefunctions

To apply boundary conditions at $x = 0$ and $x = L$, we require not only the wavefunctions but also their spatial derivatives. We denote the derivative of $\Psi(x)$ with respect to x by $\Psi'(x) = \frac{d\Psi(x)}{dx}$.

Taking the derivative of the wavefunction in Region I (Eq. 10) with respect to x , we obtain:

$$\Psi'_I(x) = \frac{d\Psi_I(x)}{dx} = ikAe^{ikx} - ikBe^{-ikx} \quad (21)$$

Taking the derivative of the wavefunction in Region II (Eq. 15) with respect to x , we get:

$$\Psi'_{II}(x) = \frac{d\Psi_{II}(x)}{dx} = qCe^{qx} - qDe^{-qx} \quad (22)$$

Taking the derivative of the wavefunction in Region III (Eq. 20) with respect to x , we find:

$$\Psi'_{III}(x) = \frac{d\Psi_{III}(x)}{dx} = ikFe^{ikx} \quad (23)$$

These derivatives will be used to enforce continuity of both $\Psi(x)$ and $\Psi'(x)$ at the boundaries $x = 0$ and $x = L$, ensuring a physically consistent solution across all regions.

Boundary Conditions at $x = 0$

To ensure physical continuity across the potential step at $x = 0$, both the wavefunction $\Psi(x)$ and its spatial derivative $\Psi'(x)$ must be continuous. This guarantees conservation of the probability density and a smooth transition of the quantum state.

The first boundary condition enforces the continuity of the wavefunction:

$$\Psi_I(0) = \Psi_{II}(0) \quad (24)$$

Substituting from Eqs. 10 and 15, we obtain:

$$A + B = C + D \quad (25)$$

The second boundary condition enforces continuity of the derivative of the wavefunction:

$$\Psi'_I(0) = \Psi'_{II}(0) \quad (26)$$

Substituting from Eqs. 21 and 22, we get:

$$ikA - ikB = qC - qD \quad (27)$$

Dividing Eq. 27 by ik , we isolate the difference:

$$A - B = \frac{q}{ik}(C - D) = i\alpha(D - C) \quad (28)$$

where we have defined

$$\alpha := \frac{q}{k} \quad (29)$$

Adding Eqs. 25 and 28, then dividing by 2:

$$A = \frac{1}{2}[(C + D) + i\alpha(D - C)] = \frac{1}{2}[C(1 - i\alpha) + D(1 + i\alpha)] \quad (30)$$

Subtracting Eq. 28 from Eq. 25, then dividing by 2:

$$B = \frac{1}{2}[(C + D) - i\alpha(D - C)] = \frac{1}{2}[C(1 + i\alpha) + D(1 - i\alpha)] \quad (31)$$

Boundary Conditions at $x = L$

At the second interface, $x = L$, the wavefunction transitions from the barrier region (Region II) to free space (Region III). As before, both the wavefunction $\Psi(x)$ and its spatial derivative $\Psi'(x)$ must be continuous to ensure conservation of probability density and smooth propagation.

The third boundary condition enforces continuity of the wavefunction:

$$\Psi_{II}(L) = \Psi_{III}(L) \quad (32)$$

Substituting from Eqs. 15 and 20, we obtain:

$$Ce^{qL} + De^{-qL} = Fe^{ikL} \quad (33)$$

The fourth boundary condition enforces continuity of the derivative of the wavefunction:

$$\Psi'_{II}(L) = \Psi'_{III}(L) \quad (34)$$

Substituting from Eqs. 22 and 23, we get:

$$qCe^{qL} - qDe^{-qL} = ikFe^{ikL} \quad (35)$$

Dividing Eq. 35 by q , we get:

$$Ce^{qL} - De^{-qL} = \frac{ik}{q}Fe^{ikL} = i\beta Fe^{ikL} \quad (36)$$

where we have defined

$$\beta := \frac{k}{q} = \frac{1}{\alpha} \quad (37)$$

Adding Eqs. 33 and 36, we get:

$$2Ce^{qL} = (1 + i\beta)Fe^{ikL} \quad (38)$$

Dividing by $2e^{qL}$, we get:

$$C = \frac{1 + i\beta}{2}Fe^{ikL}e^{-qL} \quad (39)$$

Similarly, subtracting Eq. 36 from Eq. 33, we get:

$$2De^{-qL} = (1 - i\beta)Fe^{ikL} \quad (40)$$

Multiplying both sides by $e^{qL}/2$, we get:

$$D = \frac{1 - i\beta}{2}Fe^{ikL}e^{qL} \quad (41)$$

A and B with respect to F

We now express the incident and reflected coefficients A and B in terms of the transmitted coefficient F . Substituting the expressions for C and D from Eqs. 39 and 41 into Eq. 30, we get

$$\begin{aligned} A &= \frac{Fe^{ikL}}{4} [(1 + i\beta)(1 - i\alpha)e^{-qL} + (1 - i\beta)(1 + i\alpha)e^{qL}] \\ &= \frac{Fe^{ikL}}{4} [(1 + i(\beta - \alpha) + \alpha\beta)e^{-qL} + (1 - i(\beta - \alpha) + \alpha\beta)e^{qL}] \end{aligned} \quad (42)$$

From Eqs. 29 and 37, we get an identity given as

$$\alpha\beta = \left(\frac{q}{k}\right) \left(\frac{k}{q}\right) = 1 \quad (43)$$

Using Equation 43 in Equation 42, we get

$$\begin{aligned}
A &= \frac{Fe^{ikL}}{4} [(2 + i(\beta - \alpha))e^{-qL} + (2 - i(\beta - \alpha))e^{qL}] \\
&= Fe^{ikL} \left[\frac{e^{-qL} + e^{qL}}{2} + \frac{i}{2}(\beta - \alpha) \left(\frac{e^{-qL} - e^{qL}}{2} \right) \right]
\end{aligned} \tag{44}$$

Definitions of Hyperbolic Functions (General Argument z)

$$\cosh(z) = \frac{e^z + e^{-z}}{2} \tag{45}$$

$$\sinh(z) = \frac{e^z - e^{-z}}{2} \tag{46}$$

Using the hyperbolic identities from Eqs. 45 and 46 in to Equation 44, we obtain

$$A = Fe^{ikL} \left[\cosh(qL) - \frac{i}{2}(\beta - \alpha) \sinh(qL) \right]. \tag{47}$$

Similarly, substituting the expressions for C and D from Eqs. 39 and 41 into Eq. 31, and proceeding with similar steps we get

$$\begin{aligned}
B &= \frac{Fe^{ikL}}{4} [(1 + i\beta)(1 + i\alpha)e^{-qL} + (1 - i\beta)(1 - i\alpha)e^{qL}] \\
&= \frac{Fe^{ikL}}{4} [(1 + i(\beta + \alpha) - \alpha\beta)e^{-qL} + (1 - i(\beta + \alpha) - \alpha\beta)e^{qL}] \\
&= \frac{Fe^{ikL}}{4} [i(\beta + \alpha)e^{-qL} - i(\beta + \alpha)e^{qL}] \\
&= \frac{iFe^{ikL}}{2}(\beta + \alpha) \left(\frac{e^{-qL} - e^{qL}}{2} \right) \\
&= -\frac{iFe^{ikL}(\beta + \alpha) \sinh(qL)}{2}.
\end{aligned} \tag{48}$$

Transmission Coefficient

Now, Equation 47 can be rewritten as

$$\frac{F}{A} = e^{-ikL} \left[\cosh(qL) - \frac{i}{2}(\beta - \alpha) \sinh(qL) \right]^{-1} \tag{49}$$

To compute the transmission probability (fraction of the incoming wave that is transmitted

through the barrier), we take the squared modulus:

$$\begin{aligned}
T = \left| \frac{F}{A} \right|^2 &= \left(\frac{F}{A} \right) \left(\frac{F}{A} \right)^* \\
&= \left[\cosh(qL) - \frac{i}{2}(\beta - \alpha) \sinh(qL) \right]^{-1} \left[\cosh(qL) + \frac{i}{2}(\beta - \alpha) \sinh(qL) \right]^{-1} \\
&= \left[\cosh^2(qL) + \frac{1}{4}(\beta - \alpha)^2 \sinh^2(qL) \right]^{-1} \\
&= \left[1 + \left(1 + \frac{1}{4}(\beta - \alpha)^2 \right) \sinh^2(qL) \right]^{-1} \\
&= [1 + \gamma \sinh^2(qL)]^{-1} = \frac{1}{1 + \gamma \sinh^2(qL)} \tag{50}
\end{aligned}$$

In the second last step, we used the hyperbolic identity:

Hyperbolic Identity
$\cosh^2(z) = 1 + \sinh^2(z) \tag{51}$

Identity: $1 + \frac{1}{4}(\beta - \alpha)^2 = \frac{1}{4}(\beta + \alpha)^2$
$ \begin{aligned} 1 + \frac{1}{4}(\beta - \alpha)^2 &= 1 + \frac{1}{4}(\beta^2 - 2\alpha\beta + \alpha^2) \\ &= \frac{1}{4}(\beta^2 + \alpha^2 + 2\alpha\beta) \end{aligned} \tag{52} $
Now, using $\alpha\beta = 1$ from Equation 43, we substitute into Equation 52:
$ \begin{aligned} 1 + \frac{1}{4}(\beta - \alpha)^2 &= \frac{1}{4}(\beta^2 + \alpha^2 + 2) \\ &= \frac{1}{4}(\beta + \alpha)^2 \end{aligned} \tag{53} $

The γ in [Equation 50](#) is given as

$$\gamma = 1 + \frac{1}{4}(\beta - \alpha)^2 = \frac{1}{4}(\beta + \alpha)^2 \quad (54)$$

$$\begin{aligned} &= 1 + \frac{1}{4} \left(\frac{k}{q} - \frac{q}{k} \right)^2 \\ &= 1 + \frac{1}{4} \left(\frac{k^2}{q^2} + \frac{q^2}{k^2} - 2 \right) \\ &= \frac{1}{4} \left(\frac{k^2}{q^2} + \frac{q^2}{k^2} + 2 \right) \end{aligned} \quad (55)$$

$$\begin{aligned} &= \frac{1}{4} \left(\frac{E}{V_0 - E} + \frac{V_0 - E}{E} + 2 \right) \\ &= \frac{V_0^2}{4E(V_0 - E)} = \frac{1}{4 \left(\frac{E}{V_0} \right) \left(1 - \frac{E}{V_0} \right)} \end{aligned} \quad (56)$$

where, in the second equality in the first line, we have used the identity in [Equation 53](#), in the second line, we have substituted expressions of α and β from Eqs. [29](#) and [37](#) respectively, and finally, in the second last line, we have substituted the definitions of k and q from [Equation 9](#) and [Equation 14](#) respectively.

Reflection Coefficient

Dividing [Equation 48](#) by [Equation 47](#), we get

$$\frac{B}{A} = \frac{-\frac{i}{2}(\beta + \alpha) \sinh(qL)}{\cosh(qL) - \frac{i}{2}(\beta - \alpha) \sinh(qL)} \quad (57)$$

To compute the reflection probability, we take the squared modulus given as

$$\begin{aligned} R = \left| \frac{B}{A} \right|^2 &= \left(\frac{B}{A} \right) \left(\frac{B}{A} \right)^* \\ &= \frac{\frac{1}{4}(\beta + \alpha)^2 \sinh^2(qL)}{\cosh^2(qL) + \frac{1}{4}(\beta - \alpha)^2 \sinh^2(qL)} \\ &= \frac{\frac{1}{4}(\beta + \alpha)^2 \sinh^2(qL)}{1 + \left(1 + \frac{1}{4}(\beta - \alpha)^2 \right) \sinh^2(qL)} \\ &= \frac{\gamma \sinh^2(qL)}{1 + \gamma \sinh^2(qL)} \end{aligned} \quad (58)$$

where in the last step we have used both the expressions of γ from [Equation 54](#).

Conservation of probability density

$$T + R = \frac{1}{1 + \gamma \sinh^2(qL)} + \frac{\gamma \sinh^2(qL)}{1 + \gamma \sinh^2(qL)} = 1 \quad (59)$$

This follows directly by adding the expressions for T from [Equation 50](#) and R from [Equation 58](#), both written in terms of γ .

The result $T + R = 1$ reflects the conservation of probability density in quantum mechanics: every incident particle must either tunnel through the barrier or be reflected by it. Since there is no absorption or loss in the potential barrier, the total probability density is conserved.

Limiting Case: Broad and High Barrier (Extended Analysis)

Let us define a small parameter ε as the inverse of the dominant denominator term:

$$\varepsilon := \frac{1}{\gamma \sinh^2(qL)} \quad (60)$$

This parameter becomes exponentially small in the limit $qL \gg 1$, and will serve as an expansion variable. Using Eq. 60, we rewrite Equation 50 and Equation 58 respectively as:

$$T = \frac{\varepsilon}{1 + \varepsilon}, \quad R = \frac{1}{1 + \varepsilon} \quad (61)$$

Now, for a small ε limit, we expand both expressions to first order:

$$T = \frac{\varepsilon}{1 + \varepsilon} \approx \varepsilon(1 - \varepsilon) \approx \varepsilon - \varepsilon^2 \approx \varepsilon \quad (62)$$

$$R = \frac{1}{1 + \varepsilon} \approx 1 - \varepsilon \quad (63)$$

Large Argument Approximation

$$\sinh(z) \approx \frac{1}{2}e^z \quad \text{for } z \gg 1 \quad (64)$$

We use the approximation in Equation 64 for $qL \gg 1$, so that

$$\sinh(qL) \approx \frac{1}{2}e^{qL} \quad (65)$$

Substituting into Eq. 60, we get:

$$\varepsilon \approx \frac{1}{\gamma \left(\frac{1}{2}e^{qL}\right)^2} = \frac{4e^{-2qL}}{\gamma} = 16 \left(\frac{E}{V_0}\right) \left(1 - \frac{E}{V_0}\right) e^{-2qL} \quad (66)$$

In the last equality, we have used the expression of γ from Eq. 56. Finally, using Eq. 66 in Eqs. 62 and 63, we get:

$$T \approx 16 \left(\frac{E}{V_0}\right) \left(1 - \frac{E}{V_0}\right) e^{-2qL} \quad (67)$$

$$R \approx 1 - 16 \left(\frac{E}{V_0}\right) \left(1 - \frac{E}{V_0}\right) e^{-2qL} \quad (68)$$

These expressions show that:

- **Transmission probability** decays exponentially with barrier width L and height V_0 , modulated by the energy ratio $\frac{E}{V_0}$.
- **Reflection probability** approaches unity, with a small correction proportional to the tunneling amplitude.

This confirms the classical intuition: for a broad and high barrier, quantum tunneling becomes vanishingly small, and nearly all particles are reflected. This can be seen more clearly below:

Extreme Limit: $qL \rightarrow \infty$ (Asymptotic Suppression)

In the extreme case where the barrier width L or height V_0 becomes arbitrarily large such that $qL \rightarrow \infty$, the exponential suppression dominates all prefactors. From Eq. 66, we have $\varepsilon \approx 16 \left(\frac{E}{V_0}\right) \left(1 - \frac{E}{V_0}\right) e^{-2qL}$. Taking the limit explicitly:

$$\lim_{qL \rightarrow \infty} \varepsilon = 0 \quad (69)$$

Substituting into Eqs. 62 and 63, we obtain:

$$\lim_{qL \rightarrow \infty} T = 0 \quad (70)$$

$$\lim_{qL \rightarrow \infty} R = 1 \quad (71)$$

Asymptotic Barrier Suppression

In the limit $qL \rightarrow \infty$, the transmission probability vanishes exponentially:

$$T \rightarrow 0, \quad R \rightarrow 1 \quad (72)$$

This reflects complete reflection and zero tunneling—an effective classical barrier.

Thus, quantum tunneling becomes **asymptotically forbidden**, and the system behaves as if the barrier were impenetrable. This limit serves as a benchmark for comparing tunneling amplitudes in realistic scenarios.

Region II revisited: Amplitude Ratio

We have $C = \frac{1+i\beta}{2} F e^{ikL} e^{-qL}$ (from Equation 39) and $D = \frac{1-i\beta}{2} F e^{ikL} e^{qL}$ (from Equation 41)

This gives:

$$C + D = \frac{1+i\beta}{2} F e^{ikL} e^{-qL} + \frac{1-i\beta}{2} F e^{ikL} e^{qL} \quad (73)$$

$$= \frac{F e^{ikL}}{2} [(1+i\beta)e^{-qL} + (1-i\beta)e^{qL}] \quad (74)$$

$$= \frac{F e^{ikL}}{2} [(e^{qL} + e^{-qL}) + i\beta(e^{-qL} - e^{qL})] \quad (75)$$

$$= F e^{ikL} [\cosh(qL) - i\beta \sinh(qL)] \quad (76)$$

In the last line, we have used the hyperbolic identities: $e^{qL} + e^{-qL} = 2 \cosh(qL)$ (from Equation 45) and $e^{-qL} - e^{qL} = -2 \sinh(qL)$ (from Equation 46).

This gives

$$|C + D| = |F| \sqrt{\cosh^2(qL) + \beta^2 \sinh^2(qL)} = |F| \sqrt{1 + (1 + \beta^2) \sinh^2(qL)} \quad (77)$$

Similarly, we have $C e^{qL} = \frac{1+i\beta}{2} F e^{ikL}$ (from Equation 38) and $D e^{-qL} = \frac{1-i\beta}{2} F e^{ikL}$ (from Equation 40). This gives

$$C e^{qL} + D e^{-qL} = \left(\frac{1+i\beta}{2} + \frac{1-i\beta}{2} \right) F e^{ikL} = F e^{ikL} \quad (78)$$

This gives

$$|Ce^{qL} + De^{-qL}| = |F| \quad (79)$$

Dividing Equation 79 by Equation 77, we get

$$\frac{|Ce^{qL} + De^{-qL}|}{|C + D|} = \frac{1}{\sqrt{1 + (1 + \beta^2) \sinh^2(qL)}} \quad (80)$$

From the definition of β in Equation 37, we know it is real. So, β^2 is always positive. The combination qL is also real and positive. This gives $\sinh^2(qL)$ is always positive. Consequently, these two conditions give:

Amplitude Decay in Region II

$$\frac{|\Psi_{II}(L)|}{|\Psi_{II}(0)|} = \frac{1}{\sqrt{1 + (1 + \beta^2) \sinh^2(qL)}} < 1 \quad (81)$$

This confirms that the wavefunction decays inside Region II.

Limiting Case: Broad and High Barrier (Amplitude Ratio Analysis)

Let us define a small parameter ε_2 as the inverse of the dominant denominator term in Equation 81:

$$\varepsilon_2 := \frac{1}{(1 + \beta^2) \sinh^2(qL)} \quad (82)$$

This parameter becomes exponentially small in the limit $qL \gg 1$, and will serve as an expansion variable. Using Eq. 82, we rewrite Eq. 81 as:

$$\frac{|\Psi_{II}(L)|}{|\Psi_{II}(0)|} = \sqrt{\frac{\varepsilon_2}{1 + \varepsilon_2}} \quad (83)$$

Now for small ε_2 , we expand the square root:

$$\sqrt{\frac{\varepsilon_2}{1 + \varepsilon_2}} \approx \sqrt{\varepsilon_2} \left(1 - \frac{\varepsilon_2}{2}\right) \approx \sqrt{\varepsilon_2} \quad \text{to leading order} \quad (84)$$

Using Equation 64, we have large argument approximation as $\sinh(qL) \approx \frac{1}{2}e^{qL}$ for $qL \gg 1$. Substituting this into Eq. 82, we get:

$$\varepsilon_2 \approx \frac{1}{(1 + \beta^2) \left(\frac{1}{2}e^{qL}\right)^2} = \frac{4e^{-2qL}}{1 + \beta^2} \quad (85)$$

Finally, using Eq. 85 in Eq. 84, we obtain:

$$\frac{|\Psi_{II}(L)|}{|\Psi_{II}(0)|} \approx \sqrt{\frac{4e^{-2qL}}{1 + \beta^2}} = \frac{2e^{-qL}}{\sqrt{1 + \beta^2}} \quad (86)$$

Exponential Suppression of Amplitude ($qL \gg 1$)

In the broad and high barrier limit $qL \gg 1$, the amplitude ratio decays exponentially:

$$\frac{|\Psi_{II}(L)|}{|\Psi_{II}(0)|} \approx \frac{2e^{-qL}}{\sqrt{1 + \beta^2}} \quad (87)$$

This confirms that the wavefunction is strongly suppressed inside Region II for $qL \gg 1$.

Extreme Limit: $qL \rightarrow \infty$ (Asymptotic Amplitude Suppression)

In the extreme case where the barrier width L or height V_0 becomes arbitrarily large such that $qL \rightarrow \infty$, we have from Eq. 85 as:

$$\lim_{qL \rightarrow \infty} \varepsilon_2 = 0 \quad (88)$$

Substituting into Eq. 83, we obtain:

$$\lim_{qL \rightarrow \infty} \frac{|\Psi_{II}(L)|}{|\Psi_{II}(0)|} = \lim_{\varepsilon_2 \rightarrow 0} \sqrt{\frac{\varepsilon_2}{1 + \varepsilon_2}} = 0 \quad (89)$$

Asymptotic Amplitude Suppression ($qL \rightarrow \infty$)

In the limit $qL \rightarrow \infty$, the amplitude ratio vanishes exponentially:

$$\frac{|\Psi_{II}(L)|}{|\Psi_{II}(0)|} \rightarrow 0 \quad (90)$$

This reflects complete suppression of the wavefunction inside the barrier for $qL \rightarrow \infty$.

Thus, in the extreme barrier limit, the wavefunction becomes exponentially localized near the entrance of Region II, and the probability of finding the particle deeper into the barrier becomes vanishingly small.

Physical Interpretations through the Probability Current Density

Recap of the Probability Current Density

We begin with the time-dependent Schrödinger equation in a general dimension:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V(\vec{r})\psi \quad (91)$$

Taking the complex conjugate, we obtain:

$$-i\hbar \frac{\partial \psi^*}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi^* + V(\vec{r})\psi^* \quad (92)$$

Note that, here we assume that $V(\vec{r})$ is real.

Now multiply Eq. 91 by ψ^* and Eq. 92 by ψ , then subtract:

$$\psi^* \left(i\hbar \frac{\partial \psi}{\partial t} \right) - \psi \left(-i\hbar \frac{\partial \psi^*}{\partial t} \right) = \psi^* \left(-\frac{\hbar^2}{2m} \nabla^2 \psi \right) - \psi \left(-\frac{\hbar^2}{2m} \nabla^2 \psi^* \right) \quad (93)$$

Simplifying the left-hand side:

$$i\hbar \left(\psi^* \frac{\partial \psi}{\partial t} + \psi \frac{\partial \psi^*}{\partial t} \right) = i\hbar \frac{\partial}{\partial t} (\psi^* \psi) = i\hbar \frac{\partial \rho}{\partial t} \quad (94)$$

where $\rho = \psi^* \psi = |\psi|^2$ is the probability density.

So Eq. 93 becomes:

$$i\hbar \frac{\partial \rho}{\partial t} = -\frac{\hbar^2}{2m} (\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*) \quad (95)$$

Simplifying the Spatial Term

We now simplify the right-hand side using the identity:

$$\nabla^2 f = \vec{\nabla} \cdot \vec{\nabla} f \quad (96)$$

So:

$$w := \psi^* \nabla^2 \psi - \psi \nabla^2 \psi^* = \psi^* \vec{\nabla} \cdot \vec{\nabla} \psi - \psi \vec{\nabla} \cdot \vec{\nabla} \psi^* \quad (97)$$

Let us define: $\vec{U}_1 := \vec{\nabla} \psi$ and $\vec{U}_2 := \vec{\nabla} \psi^*$ So:

$$w = \psi^* \vec{\nabla} \cdot \vec{U}_1 - \psi \vec{\nabla} \cdot \vec{U}_2 \quad (98)$$

The product rule for divergence:

Divergence of Scalar–Vector Product

$$\vec{\nabla} \cdot (s\vec{U}) = s(\vec{\nabla} \cdot \vec{U}) + (\vec{\nabla} s) \cdot \vec{U} \quad (99)$$

The product rule for divergence is rewritten as

$$s(\vec{\nabla} \cdot \vec{U}) = \vec{\nabla} \cdot (s\vec{U}) - (\vec{\nabla} s) \cdot \vec{U} \quad (100)$$

Now apply Eq. 100 to each term of w :

$$\psi^* \vec{\nabla} \cdot \vec{U}_1 = \vec{\nabla} \cdot (\psi^* \vec{U}_1) - (\vec{\nabla} \psi^*) \cdot \vec{U}_1 = \vec{\nabla} \cdot (\psi^* \vec{\nabla} \psi) - (\vec{\nabla} \psi^*) \cdot (\vec{\nabla} \psi) \quad (101)$$

$$\psi \vec{\nabla} \cdot \vec{U}_2 = \vec{\nabla} \cdot (\psi \vec{U}_2) - (\vec{\nabla} \psi) \cdot \vec{U}_2 = \vec{\nabla} \cdot (\psi \vec{\nabla} \psi^*) - (\vec{\nabla} \psi) \cdot (\vec{\nabla} \psi^*) \quad (102)$$

Subtracting the two:

$$w = \left[\vec{\nabla} \cdot (\psi^* \vec{\nabla} \psi) - (\vec{\nabla} \psi^*) \cdot (\vec{\nabla} \psi) \right] - \left[\vec{\nabla} \cdot (\psi \vec{\nabla} \psi^*) - (\vec{\nabla} \psi) \cdot (\vec{\nabla} \psi^*) \right] \quad (103)$$

$$= \vec{\nabla} \cdot (\psi^* \vec{\nabla} \psi - \psi \vec{\nabla} \psi^*) \quad (104)$$

Substitute back into Eq. 95:

$$i\hbar \frac{\partial \rho}{\partial t} = -\frac{\hbar^2}{2m} \vec{\nabla} \cdot (\psi^* \vec{\nabla} \psi - \psi \vec{\nabla} \psi^*) \quad (105)$$

Divide both sides by $i\hbar$:

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \left[\frac{\hbar}{2mi} (\psi^* \vec{\nabla} \psi - \psi \vec{\nabla} \psi^*) \right] = 0 \quad (106)$$

Defining the Probability Current Density

We define the probability current density vector $\vec{J}$ as:

$$\vec{J} := \frac{\hbar}{2mi} \left(\psi^* \vec{\nabla} \psi - \psi \vec{\nabla} \psi^* \right) \quad (107)$$

Alternatively, using the identity $\text{Im}(f) = \frac{1}{2i}(f - f^*)$, we write:

$$\vec{J} = \frac{\hbar}{m} \text{Im} \left(\psi^* \vec{\nabla} \psi \right) \quad (108)$$

Final Continuity Equation and Interpretation

Substituting Eq. 107 into Eq. 106, we obtain the continuity equation:

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{J} = 0 \quad (109)$$

Physical Interpretation

The equation ensures that the probability density cannot be created or destroyed; it can only change within a region through divergence (outflow) or convergence (inflow) of the probability current density across the region's boundaries.

Special Case: Time-Independent Schrödinger Equation in 1D

We now consider the case where the wavefunction separates into spatial and temporal parts:

$$\psi(x, t) = \Psi(x) e^{-iEt/\hbar} \quad (110)$$

Here, $\Psi(x)$ is the spatial part and $e^{-iEt/\hbar}$ encodes the time evolution. Consequently:

$$|\psi(x, t)|^2 = |\Psi(x)|^2 \quad (111)$$

The probability density becomes time-independent:

$$\rho(x, t) = |\psi(x, t)|^2 = |\Psi(x)|^2 = \rho(x) \quad (112)$$

Therefore, its time derivative vanishes:

$$\frac{\partial \rho}{\partial t} = 0 \quad (113)$$

From Equation 109, we have the continuity equation: $\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{J} = 0$. Using the above equation it, we conclude:

$$\vec{\nabla} \cdot \vec{J} = 0 \quad (114)$$

Expression for $J(x)$ in 1D

Recall the general definition of the probability current density in [Equation 108](#) as $\vec{J} = \frac{\hbar}{m} \text{Im}(\psi^* \vec{\nabla} \psi)$. For a time-harmonic (corresponding to the time-independent Schrödinger equation) state $\psi(x, t) = \Psi(x)e^{-iEt/\hbar}$ in 1D, we have

$$\vec{\nabla} \psi = \Psi'(x)e^{-iEt/\hbar} \hat{x}, \quad \psi^* = \Psi^*(x)e^{iEt/\hbar}, \quad (115)$$

Using these, we get:

$$\vec{J}(x) = J(x) \hat{x} \quad (116)$$

$$|\vec{J}(x)| = J(x) \equiv J_x(x) = \frac{\hbar}{m} \text{Im}[\Psi^*(x)\Psi'(x)], \quad (117)$$

Putting this in [Equation 114](#), we get

$$\nabla \cdot \vec{J} = \frac{dJ}{dx} = 0 \quad \Rightarrow \quad J(x) = \text{constant (in space)}. \quad (118)$$

Physical interpretation in 1D (x-direction)

For a time-independent (stationary) state $\psi(x, t) = \Psi(x)e^{-iEt/\hbar}$:

- The probability density $\rho(x) = |\Psi(x)|^2$ is time-independent.
- The probability current density has only an x -component,

$$J(x) = \frac{\hbar}{m} \text{Im}[\Psi^*(x)\Psi'(x)], \quad \vec{J}(x) = J(x) \hat{x}.$$

- Stationarity implies $\partial_x J = 0$, so J is spatially constant. Physically: there is no local accumulation or depletion of probability density; probability density may flow steadily through the system at a constant rate, but the local density at each point remains unchanged in time.

Application in This Case: Probability Current in Each Region

Region I ($x < 0$): Free Space

We begin with the wavefunction $\Psi_I(x) = Ae^{ikx} + Be^{-ikx}$ from [Equation 10](#). Its complex conjugate is:

$$\Psi_I^*(x) = A^*e^{-ikx} + B^*e^{ikx} \quad (119)$$

And its spatial derivative is: $\Psi_I'(x) = ikAe^{ikx} - ikBe^{-ikx}$ from [Equation 21](#). Using this and the above equation, we compute the product $\Psi_I^*(x)\Psi_I'(x)$ as:

$$\Psi_I^*(x)\Psi_I'(x) = (A^*e^{-ikx} + B^*e^{ikx})(ikAe^{ikx} - ikBe^{-ikx}) \quad (120)$$

$$= ikA^*A - ikB^*B - ikA^*Be^{-2ikx} + ikAB^*e^{2ikx} \quad (121)$$

Euler Identities for Imaginary Arguments

$$e^{iz} = \cos(z) + i \sin(z) \quad (122)$$

$$e^{-iz} = \cos(z) - i \sin(z), \quad (123)$$

Expanding the exponentials using Euler's formulas from [Equation 122](#) and [Equation 123](#) as: $e^{2ikx} = \cos(2kx) + i \sin(2kx)$ and $e^{-2ikx} = \cos(2kx) - i \sin(2kx)$ respectively. Using these, we get:

$$-ikA^*Be^{-2ikx} = -ikA^*B \cos(2kx) + kA^*B \sin(2kx), \quad (124)$$

$$+ikAB^*e^{2ikx} = +ikAB^* \cos(2kx) - kAB^* \sin(2kx) \quad (125)$$

Combining these:

$$-ikA^*Be^{-2ikx} + ikAB^*e^{2ikx} = ik(AB^* - A^*B) \cos(2kx) - k(A^*B + AB^*) \sin(2kx) \quad (126)$$

Therefore:

$$\Psi_I^*(x)\Psi_I'(x) = ikA^*A - ikB^*B + ik(AB^* - A^*B) \cos(2kx) - k(A^*B + AB^*) \sin(2kx) \quad (127)$$

We now express A and B in terms of their real and imaginary parts:

$$A = a_r + ia_i, \quad B = b_r + ib_i \quad (128)$$

Then the product AB^* becomes:

$$AB^* = (a_r + ia_i)(b_r - ib_i) = a_rb_r + a_ib_i + i(a_ib_r - a_rb_i) \quad (129)$$

So the real and imaginary parts of AB^* are:

$$\text{Re}(AB^*) = a_rb_r + a_ib_i, \quad \text{Im}(AB^*) = a_ib_r - a_rb_i \quad (130)$$

Similarly, we compute A^*B :

$$A^*B = (a_r - ia_i)(b_r + ib_i) = a_rb_r + a_ib_i + i(a_rb_i - a_ib_r) \quad (131)$$

Now we compute the sum and difference:

$$AB^* + A^*B = 2(a_rb_r + a_ib_i) = 2\text{Re}(AB^*) \quad (132)$$

$$AB^* - A^*B = 2i(a_ib_r - a_rb_i) = 2i\text{Im}(AB^*) \quad (133)$$

We summarize these in [Equation 134](#) and [Equation 135](#) respectively.

Complex Identities

For any complex numbers A and B , we have:

$$AB^* + A^*B = 2\text{Re}(AB^*) \quad (134)$$

$$AB^* - A^*B = 2i\text{Im}(AB^*) \quad (135)$$

Substituting into the expression for $\Psi_I^*(x)\Psi_I'(x)$:

$$\Psi_I^*(x)\Psi_I'(x) = ik(A^*A - B^*B) + ik[2i\text{Im}(AB^*)]\cos(2kx) - 2k\text{Re}(AB^*)\sin(2kx) \quad (136)$$

$$= ik(|A|^2 - |B|^2) - 2k\text{Im}(AB^*)\cos(2kx) - 2k\text{Re}(AB^*)\sin(2kx) \quad (137)$$

Taking the imaginary part:

$$\text{Im}[\Psi_I^*(x)\Psi_I'(x)] = k(|A|^2 - |B|^2) \quad (138)$$

So the probability current density is:

$$\vec{J}_I(x) = \frac{\hbar k}{m} (|A|^2 - |B|^2) \hat{x} \quad (139)$$

Alternative derivation: Probability Current Density via Individual Components

We can compute the probability current density in Region I more alternatively by treating the incident and reflected wave components separately. Since the cross terms are real, they do not contribute to the imaginary part of $\Psi^*(x)\Psi'(x)$, which determines the current.

Incident Wave Component

$$\Psi_{\text{inc}}(x) = Ae^{ikx} \quad (140)$$

$$\Psi_{\text{inc}}^*(x) = A^*e^{-ikx}, \quad \Psi_{\text{inc}}'(x) = ikAe^{ikx} \quad (141)$$

$$\Psi_{\text{inc}}^*(x)\Psi_{\text{inc}}'(x) = A^*e^{-ikx} \cdot ikAe^{ikx} = ikA^*A = ik|A|^2 \quad (142)$$

$$\text{Im}[\Psi_{\text{inc}}^*(x)\Psi_{\text{inc}}'(x)] = k|A|^2 \quad (143)$$

$$\vec{J}_{\text{inc}}(x) = \frac{\hbar k}{m} |A|^2 \hat{x} \quad (144)$$

Reflected Wave Component

$$\Psi_{\text{ref}}(x) = Be^{-ikx} \quad (145)$$

$$\Psi_{\text{ref}}^*(x) = B^*e^{ikx}, \quad \Psi_{\text{ref}}'(x) = -ikBe^{-ikx} \quad (146)$$

$$\Psi_{\text{ref}}^*(x)\Psi_{\text{ref}}'(x) = B^*e^{ikx} \cdot (-ikBe^{-ikx}) = -ikB^*B = -ik|B|^2 \quad (147)$$

$$\text{Im}[\Psi_{\text{ref}}^*(x)\Psi_{\text{ref}}'(x)] = -k|B|^2 \quad (148)$$

$$\vec{J}_{\text{ref}}(x) = -\frac{\hbar k}{m}|B|^2 \hat{x} \quad (149)$$

Total Probability Current Density

$$\vec{J}_I(x) = \vec{J}_{\text{inc}}(x) + \vec{J}_{\text{ref}}(x) = \frac{\hbar k}{m}(|A|^2 - |B|^2) \hat{x} \quad (150)$$

The total probability current density can be rewritten w.r.t the reflection fraction R as

$$\begin{aligned} \vec{J}_I(x) &= \frac{\hbar k}{m}|A|^2 \left(1 - \left|\frac{B}{A}\right|^2\right) \hat{x} = \frac{\hbar k}{m}|A|^2 (1 - R) \hat{x} \\ &= (1 - R) \vec{J}_{\text{inc}}(x) \end{aligned} \quad (151)$$

$$= T \vec{J}_{\text{inc}}(x) \quad (152)$$

Region II ($0 \leq x \leq L$): Barrier Region

The wave function in Region II is: $\Psi_{II}(x) = Ce^{qx} + De^{-qx}$ (from [Equation 15](#)). Its complex conjugate is:

$$\Psi_{II}^*(x) = C^* e^{qx} + D^* e^{-qx} \quad (153)$$

The spatial derivative is: $\Psi'_{II}(x) = \frac{d\Psi_{II}}{dx} = qCe^{qx} - qDe^{-qx}$ (from [Equation 22](#)). Using this and the above equation, we compute the product $\Psi_{II}^*(x)\Psi'_{II}(x)$:

$$\Psi_{II}^*(x)\Psi'_{II}(x) = (C^* e^{qx} + D^* e^{-qx}) (qCe^{qx} - qDe^{-qx}) \quad (154)$$

$$= qC^* C e^{2qx} - qD^* D e^{-2qx} + qCD^* - qC^* D \quad (155)$$

Using the identity in [Equation 135](#):

$$CD^* - C^* D = 2i \text{Im}(CD^*) \quad (156)$$

This gives

$$\Psi_{II}^*(x)\Psi'_{II}(x) = qC^* C e^{2qx} - qD^* D e^{-2qx} + 2iq \text{Im}(CD^*) \quad (157)$$

Taking the imaginary part:

$$\text{Im}[\Psi_{II}^*(x)\Psi'_{II}(x)] = 2q \text{Im}(CD^*) \quad (158)$$

Therefore, the probability current density in Region II is:

$$\vec{J}_{II}(x) = \frac{2q\hbar}{m} \text{Im}(CD^*) \hat{x} \quad (159)$$

We have $C = \frac{1+i\beta}{2} F e^{ikL} e^{-qL}$ (from [Equation 39](#)) and $D = \frac{1-i\beta}{2} F e^{ikL} e^{qL}$ (from [Equation 41](#)). The complex conjugate of D is given by

$$D^* = \left(\frac{1 - i\beta}{2} F e^{ikL} e^{qL} \right)^* = \frac{1 + i\beta}{2} F^* e^{-ikL} e^{qL} \quad (160)$$

Multiplying for CD^* as:

$$CD^* = \left(\frac{1 + i\beta}{2} F e^{ikL} e^{-qL} \right) \left(\frac{1 + i\beta}{2} F^* e^{-ikL} e^{qL} \right) \quad (161)$$

$$= \frac{(1 + i\beta)^2}{4} |F|^2 = \frac{1 + 2i\beta - \beta^2}{4} |F|^2 \quad (162)$$

Taking the imaginary part, we have

$$\text{Im}(CD^*) = \frac{1}{4} \cdot 2\beta |F|^2 = \frac{\beta}{2} |F|^2 \quad (163)$$

Substituting this in [Equation 159](#):

$$\vec{J}_{II}(x) = \frac{2q\hbar}{m} \cdot \frac{\beta}{2} |F|^2 \hat{x} = \frac{q\hbar}{m} \beta |F|^2 \hat{x} \quad (164)$$

Using $\beta = \frac{k}{q}$ (from [Equation 37](#)), we get

$$\vec{J}_{II}(x) = \frac{k\hbar}{m} |F|^2 \hat{x} = \frac{\hbar k}{m} |A|^2 \left| \frac{F}{A} \right|^2 \hat{x} = T \vec{J}_{\text{inc}}(x) \quad (165)$$

—

Region III ($x > L$): Free Space Again

The wavefunction is: $\Psi_{III}(x) = F e^{ikx}$ (from [Equation 20](#)). The complex conjugate:

$$\Psi_{III}^*(x) = F^* e^{-ikx} \quad (166)$$

The spatial derivative is: $\Psi'_{III}(x) = \frac{d\Psi_{III}}{dx} = ik F e^{ikx}$ (from [Equation 23](#)). Using this and the above equation, we compute the product:

$$\Psi_{III}^* \Psi'_{III} = ik F^* F \quad (167)$$

The imaginary part is:

$$\text{Im}(\Psi_{III}^* \Psi'_{III}) = k |F|^2 \quad (168)$$

So the probability current density is:

$$\vec{J}_{III}(x) = \frac{\hbar k}{m} |F|^2 \hat{x} = \frac{\hbar k}{m} |A|^2 \left| \frac{F}{A} \right|^2 \hat{x} = T \vec{J}_{\text{inc}}(x) \quad (169)$$

Now, we can see that we have got

$$\vec{J}_I(x) = \vec{J}_{II}(x) = \vec{J}_{III}(x) \quad (170)$$

The result shows that the probability current density $\vec{J}(x)$ is the same constant vector in all three regions of the tunneling setup. Physically, this expresses the conservation of probability density flow: although the wavefunction takes different forms in each region (oscillatory outside the barrier and exponentially decaying inside), the flow of probability density, described by the probability current

density, is continuous and unbroken across the entire system. No probability density is created or destroyed inside the barrier — the incoming probability current density from the left equals the sum of the reflected and transmitted probability current densities. This is mathematically equivalent for the relation $R + T = 1$.